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Multi-Objective Task Scheduling in the Cloud Environment using Reinforcement Learning and Whale Optimization Algorithm

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***Abstract*:** Cloud Computing (CC) enables organizations and individuals to alter and revolutionize the procedure by which people obtain and utilize resources. Due to the growth of the CC, the complexity increases and the cloud systems demand an effective solution for optimal resource allocation. However, effective utilization of resources in CC systems remains a significant problem due to their scale, heterogeneity, and dynamic nature. Artificial Intelligence (AI) has grown up as an effective means to improve management of resources effectively. Task scheduling is one of the major aspects in resource allocation and this study focuses on dynamic task scheduling algorithms to effectively allocate the task to the appropriate Virtual Machines (VMs). This study examines and suggests the reinforcement learning approach with a whale optimization algorithm (RL-WOA) for efficient task allocation in the cloud environment and its performance in terms of efficiency measures in terms of QoS metrics, makespan , CPU utilization and memory usage. The implementation performed using a CloudSim simulator for demonstrating the efficiency of the suggested model

Keywords: Cloud Computing (CC), Resource Allocation, Task Scheduling(TS), Reinforcement Learning, Whale Optimization Algorithm

I.INTRODUCTION

CC is well-known as a type of distributed computing system connected by a very fast network. It offers infrastructure and applications delivered as services over the Internet. The infrastructure and execution platform provide users with dynamic assistance [1].

In CC, optimum resource allocation and task execution time between virtual machines (VM's) are critical to meeting customers' needs to process requests fast and offer quality services. One of the major issues that cloud service providers confront is effectively managing resources through physical infrastructure [2]. Cloud infrastructures are made up of data centers, hosts, and VMs, and proper work scheduling is critical for top performance. Effective scheduling is essential for saving time, money, energy, and reaction time [3].

Resource allocation influences productivity, expenses, outcomes, and SLAs. Users and providers can assign cloud resources based on workloads by utilizing flexibility, expansion, and on-demand delivery. IT costs and efficiency in operation have altered as a result of rapid and dynamic resource allocation. Identifying cloud resource management issues and potential requires a proactive rather than reactive approach to resource allocation. Reactive techniques provide resources only when there are shortages or surpluses on demand.

Organizations may need various provisioning tools to efficiently manage, customize, and use their cloud resources. When workloads are deployed across various cloud platforms, a centralized portal is created to monitor and manage all resources to ensure optimal allocation. It leads to a more optimal and efficient use of resources. One of the industry's best practices for optimal resource allocation is the task scheduling. It is a technique to distribute the incoming network traffic among numerous servers to guarantee that no server is overloaded. This improves the performance and dependability of the applications. The task scheduling classification is shown in figure 1.

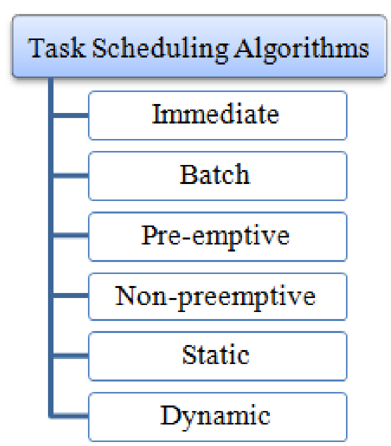

Figure 1. Types of TS Algorithms

The immediate scheduling happens whenever the tasks arrive and batch scheduling does the jobs in batches or groups. The static scheduling is performed using the previous information and the dynamic scheduling is done with the current state of the system. Pre-emptive scheduling happens by interrupting the running tasks and in non-preemption, the VM.s are allocated after finishing the currently running tasks. This study focuses on the dynamic task scheduling strategy using the reinforcement learning and whale optimization algorithm (RL-WOA). The major contributions of the study includes (i) suggesting RL approach with WOA for task scheduling for VMs in cloud. (ii) Evaluating the performance using QoS metrics makespan, CPU utilization, and memory usage

II. RELATED WORKS

Various approaches were studied in the literature for tasks scheduling in the cloud settings.. Deep Q-learning network approach is used for scheduling problems by searching for the best resource for the tasks. The method employs a randomly generated workload and the performance is evaluated using the HP2CN and NASA workloads [4].

The suggested technique in [5] DMOTS-DRL(Dynamic Multi-objective task scheduling-Deep reinforcement Learning), merges Dueling deep Q-networks and flexible prioritised replay of experiences to improve two essential goals: completion time (makespan) and energy consumption. The efficiency of DMOTS-DRL is tested using CloudSim and compared to numerous cutting-edge TS algorithms. The experimental results reveal that DMOTS-DRL beats competing algorithms in terms of makespan, energy usage and other metrics, confirming its efficacy and dependability for cloud computing services.

The study in [6] uses the Deep CNN with Fuzzy (DCNN-F) approach to differentiate cloud nodes. The complex nature of scheduling workflows in the cloud is reduced through effective learning, while energy and time usage are carefully managed. The DCNN-F is trained using cloud-based assets, and scheduling issues have been addressed with learning data. The network is iteratively tweaked and optimized via the DCNN-F feedback mechanism. By combining DCNN-F capabilities with effective resource allocation methods, researchers can improve energy and time scheduling for precedence-constrained applications in CC environments.

Table 1 Recent studies in the literature for Resource Management in Cloud

Author &year	Methodology	Purpose	Score
Sandhu et al., 2024[2]	WOA	To reduce total execution cost, total execution time Energy Consumption and overall response time	Overall System effectiveness
Alizadeh et al., 2023[3]	CDDQL Algorithm and PSO	Optimal allocation of resources by prioritizing the tasks	Makespan , response time, task completion, resource and energy usage
Mangalam palli et al., 2024[4]	DRL	Automatically arrives a decision based on the workload and the available resources	Makespan SLS violation percentage Energy consumption
Chraibi et al., 2023[5]	DMOTS – DRL	Optimization	Makespan, power consumption , degree of imbalance, resource utilization Average waiting time
Rajendran et al.,2024[6]	DCNN-F	Self adaptive DL	Energy and Time
Kamble et al.2023 [7]	CNN, LSTM, Transformer model	Predicts resources using historical data	Resource utilization , Cost effectiveness

Table 1 lists some of the recent studies found in the literature for TS cloud context.. This study contributes to the area of task scheduling in CC with the RL approach.

III.METHODOLOGY

The following multi-objective model is used for task allocation and it includes the QoS metrics like Response time (RT), Throughput (T) and Availability (A).

(i) Response Time(RT)
RT pertains to how long it takes for a request to respond to a particular type of request. It is determined by load intensity, which can be measured as the number of requests received at once or arrival rates (requests per second). QoS considers both the average RT and the RT percentile..

$$Response\ Time(RT) = \frac{TRT}{N} \quad (1)$$

In equation (1) TRT denotes the average RT and N is the total number of requests

(ii)Throughput
The amount of throughput of a service is the maximum pace at which requests may be processed. QoS metrics include functions that show the relationship between throughput and load intensity, as well as the ideal throughput.

$$Throughput(T) = \frac{N}{T} \quad (2)$$

In equation (2) T is the total time interval
(iii) Availability

The accessibility of a resource is the percentage of time it is operational and accessible to users. It is commonly referred to as the uptime %. High availability is essential for services that are critical because it lowers downtime and service interruptions while offering an impeccable user experience.

$$Availability(A) = \frac{TUT}{TUT+TDT} * 100 \qquad (3)$$

In equation (3) TUT is the total uploading time and TDT is the total down time .The tasks are assigned based on these parameters. The following objective function servers as a model for efficient task allocation.

$$Objective\ function = \max\,(RT, T, A) \qquad (4)$$

In figure 2, the tasks are allocated to the VMs , in turn it gets allocated to the corresponding physical machines(PMs).

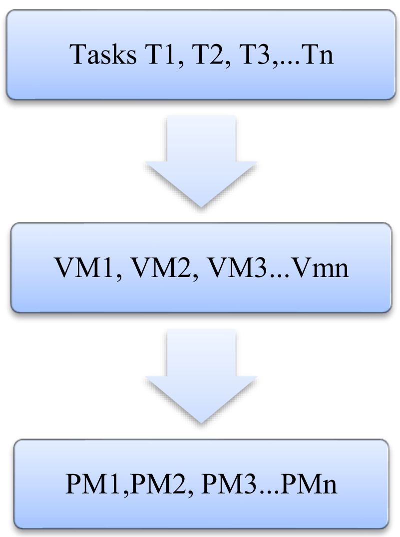

Figure 2: Task allocation to the VMs

The load for each VM is computed using the RL-WOA , CNN and LSTM algorithms .

A. Reinforcement Learning

In reinforcement learning (RL), an agent learns how to accomplish a goal by observing an unfamiliar environment. Through error and trial, RL, a branch of ML, allows AI-based systems to maximize group rewards using feedback for specific actions in a dynamic environment.

According to the principle of RL maximizing anticipated cumulative reward can specify any aim. The agent must develop the ability to identify and alter the state of the environment with its behaviors in order to maximize reward. In a federated cloud setting, scheduling real-time work is essentially a continuous choice process that RL can convert into a Markov or Semi-Markov decision problem. Three main elements make up RL: (i) the agent (the agent that learns), (ii) the setting (the environment that the agent interacts with), and (iii) the behaviors.

Through interaction and rewards for its behaviours, an agent gains knowledge about its surroundings. It shows the scheduling problem as a triplet (S, A, R), where S stands for the state, A for the action, and R for the reward[8,9].

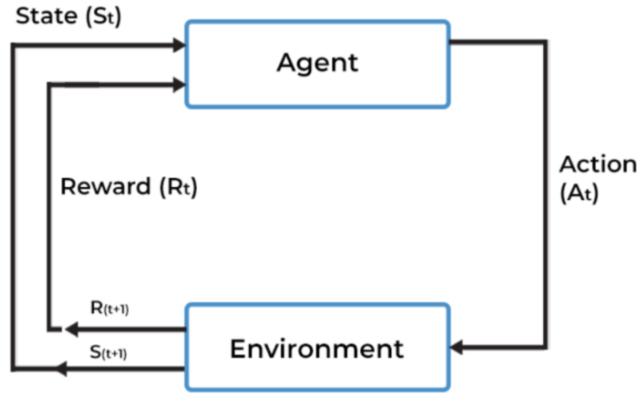

Figure 3: RL Model

Figure 3 shows an agent in a specific state (St) . It carries out an action (At) in a setting in order to accomplish a particular objective. After finishing the task, the agent is rewarded or penalized with feedback. The RL workflow, which includes agent training while accounting for the following crucial elements, is shown in Figure 4.:

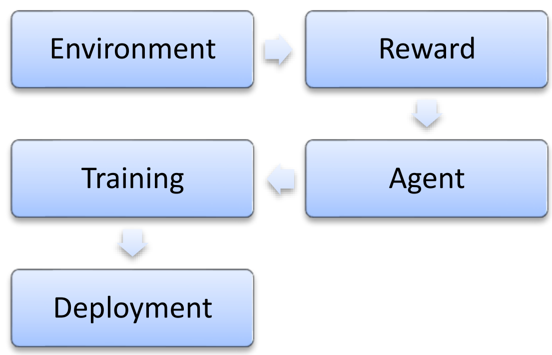

Figure 4. RL Work flow

Step I: Create an Environment

The first step in the RL approach is to decide on the environment in which the agent will continue to function. The word "environment" can refer to a simulation or a real-world system.

Step 2: Specify the award

The next stage is to define the agent's reward. It serves as a performance metric for the agent, allowing it to assess task quality versus its objectives. Furthermore, delivering suitable rewards to the agent can necessitate several iterations to choose the best one for a certain behavior.

Step 3: Define the agent.

It's possible to next construct the agent, which specifies all of the relevant rules, including the RL training algorithm, after the environment and rewards have been established. The following steps could be included in the process:

- Employ suitable neural networks to indicate the policy.
- Select an appropriate RL training technique

This study uses SARSA method i.e . The State-Action-Reward-State-Action algorithm is a policy-based technique. Thus, it does not follow Q-learning's greedy strategy. Instead, SARSA learns from its existing condition and takes actions to implement the RL process.

Step IV: Train and validate the agent.

Improve the training policy by training and validating the agent. Also, focus on the incentive structure, RL's policy architecture, and the training process. RL training might take anywhere from a few minutes to several days, depending on the final application. Thus, for a complex set of applications, faster training is achieved by using a system design that includes multiple CPUs, GPUs, and computation operations simultaneously.

Step V: Policy Implementation

When optimal judgments or results are not reached while adopting these policies, it is occasionally necessary to return to the starting stages of the RL workflow. The proposed workflow is shown in figure 5.

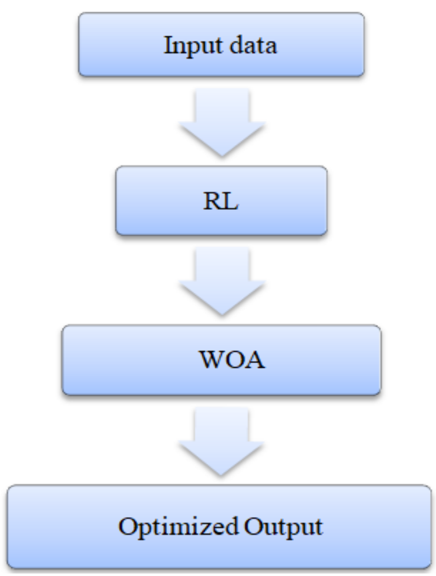

Figure 5. Proposed RL and WOA model for task allocation

B. Whale Optimization Algorithm (WOA)

WOA is a nature-inspired approach that imitates humpback whales' hunting patterns. It consists of three primary steps: encircling the target, launching a bubble-net attack, and looking for prey. The algorithm begins with an unknown location for the ideal solution and gradually updates it through encircling and spiral updating positions.

(i)Encircling prey:

Whales can locate and encircle their prey. However, in the search space, the optimal site is unknown during the early stage. In WOA, the current optimal or preferred location is surrounded by the optimal location or prey.

(ii) Bubble net attack:

After identifying the target's location, the method seeks to encircle the area around the optimal solution, an approach known as "shrinking encircling". The distance between the current location and the newly circled locations is calculated and used to update position. This is known as "spiral updating position".

(iii) Search for Prey:

Whales continue to investigate even while preying. In the meantime, individual whales seek for prey based on their location of their partner whales. In algorithmic terms, the initial step is to determine the whale population, followed by finding the whale with the greatest degree of fitness and continuing until the desired results are accomplished or an iterative limit is attained [10.11]. Initially, the multi-objective model calculates RT, throughput, and resource availability. The recommended TS algorithm, when combined with the WOA can optimally allocate work to VMs.

IV. FINDINGS AND DISCUSSIONS

The RL-WOA approach is compared with the other DL methods like CNN and LSTM approaches using the following metrics

(i) Makespan time

The highest finish time or the amount of time needed to allocate resources to a customer[38] .

(ii) Resource Utilization

For instance, the formula from equations 1 to 4 will assist in predicting overloaded machines by analyzing the present status of each machine.

$$\text{CPU utilization} = \frac{\sum \text{Currently\_running\_task}}{\text{Total Capacity}} * 100 \quad (1)$$

$$\text{RAM utilization} = \frac{\sum \text{RAM\_current\_utilization}}{\text{Total RAM}} * 100 \quad (2)$$

$$\text{BW\_ utilization} = \frac{\sum \text{BW\_current\_utilization}}{\text{Total RAM}} * 100 \quad (3)$$

$$\text{Storage utilization} = \frac{\sum \text{Storage\_current\_utilization}}{\text{Total\_Storage\_Capacity}} * 100 \quad (4)$$

$$\text{Resource utilization} = \sum \text{CPU}_{\text{utilization}} + \text{RAM}_{\text{utilization}} + \text{BW}_{\text{utilization}} + \text{Storage}_{\text{utilization}} \quad [5]$$

The above metrics are computed and plotted in the graphical representation in figure 6, 7,and 8

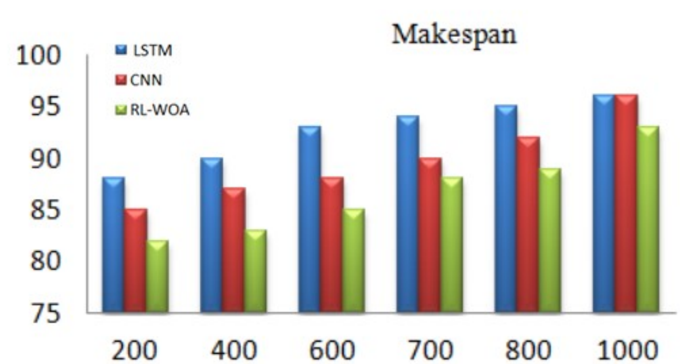

Figure 6: Makespan vs number of tasks

In figure 6, the makespan of RL-WOA is low than the other two methods. A lower makespan suggests improved resources use and work allocation.

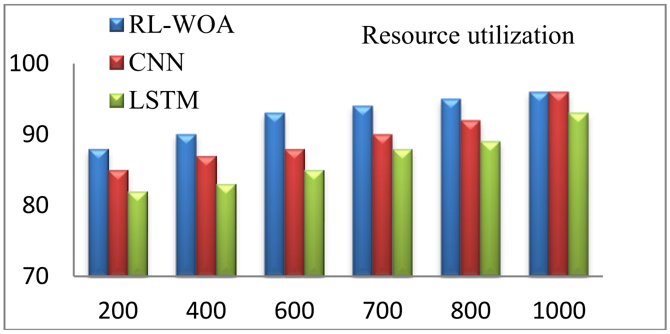

Figure 7: Resource utilization vs number of tasks

In figure 7, the resource utilization, in the case of RL-WOA is higher than the CNN and LSTM. High resource utilization requires making the best use of the resources accessible while reducing idle time.

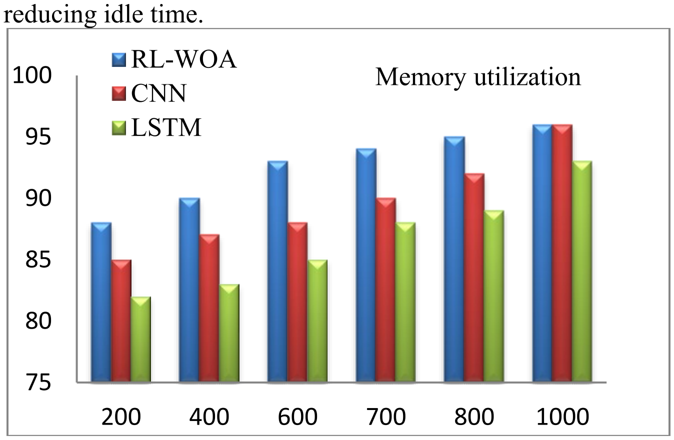

Figure 8: Memory Utilization vs number of tasks

In figure 8, the memory usage is found to be more compared to the other methods. High memory usage indicates that the system is efficiently utilizing the memory that is accessible resources without letting a substantial percentage idle. It is clear from the results that the RL-WOA outperforms the other two methods used in this study. One of the most regularly used metrics is accuracy. The accuracy is determined by the percentage of correctly categorized data points: A accuracy score close to 100% indicates that your model did not miss any true positives and can accurately distinguish between. A low accuracy score (<50%) indicates a significant number of false positives, which may be caused by an unbalanced class or untuned model hyper-parameters. A recall is basically the proportion of true positives to total positives in the real world..Precision is the proportion of real positives to total positives expected Using the following formula, metrics for accuracy, recall, and precision can be calculated. The evaluation metrics is shown in figure 9.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

$$\text{Recall}=\frac{TP}{TP+FN} \quad (2)$$

$$\text{Precision} =\frac{TP}{TP+FP} \quad (3)$$

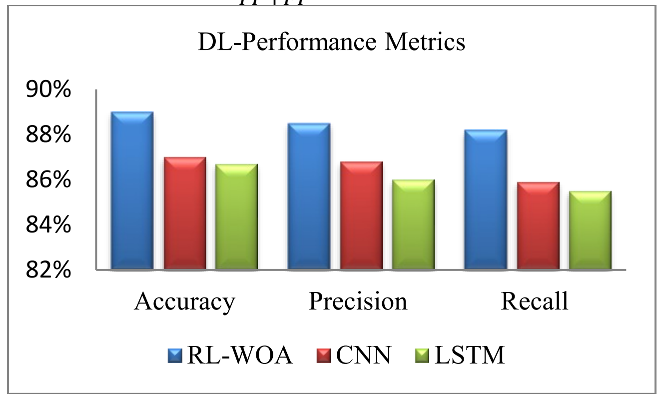

Figure 9 : Performance Metrics Vs Methods

From the figure 9, it is found that the accuracy, precision and recall for RL-WOA is found to be higher than the CNN and LSTM. RL-WOA, the learning algorithms with the optimization techniques outperforms the other two methods.

V. CONCLUSION

RL automates decision-making and learning processes. RL agents are known to learn from their surroundings and experiences without requiring close oversight or human involvement. The study examines the use of RL-WOA for task scheduling and contrasts with the CNN and LSTM. The comparison of RL-WOA to existing algorithms shows its usefulness in improving makespan, resource usage and memory usage. Moreover the RL allows for independent decision-making by learning how to maximize cumulative rewards in a training context. Therefore, through studying and exploiting, RL algorithms learn and discover optimal techniques to accomplish the intended results in a virtual environment. Further WOA approach has several advantages, including simplicity in concepts, fewer variables, and a straightforward implementation. In future, the optimized task scheduling can be accomplished by exploring the other optimization algorithms to enhance the efficiency of the resource allocation strategies

REFERENCES

1. Zhou, Guangyao, Wenhong Tian, Rajkumar Buyya, Ruini Xue, and Liang Song. "Deep reinforcement learning-based methods for resource scheduling in cloud computing: A review and future directions." Artificial Intelligence Review 57, no. 5 (2024): 124.
2. Sandhu, R., Faiz, M., Kaur, H. et al. Enhancement in performance of cloud computing task scheduling using optimization strategies. Cluster Comput 27, 6265–6288 (2024). https://doi.org/10.1007/s10586-023-04254-w
3. Alizadeh Javaheri, S.D., Ghaemi, R. & Monshizadeh Naeen, H. An autonomous architecture based on reinforcement deep neural network for resource allocation in cloud computing. Computing 106, 371–403 (2024). https://doi.org/10.1007/s00607-023-01220-7
4. Mangalampalli, S., Karri, G.R., Kumar, M. et al. DRLBTSA: Deep reinforcement learning based task-scheduling algorithm in cloud computing. Multimed Tools Appl 83, 8359–8387 (2024). https://doi.org/10.1007/s11042-023-16008-2

5. Chraibi, A., Ben Alla, S., Touhafi, A. et al. A novel dynamic multi-objective task scheduling optimization based on Dueling DQN and PER. J Supercomput 79, 21368–21423 (2023). https://doi.org/10.1007/s11227-023-05489-5
6. Rajendran, L., Shekhawat, V.S. Deep Convolutional Neural Network with a Fuzzy (DCNN-F) technique for energy and time optimized scheduling of cloud computing. Cluster Comput (2024). https://doi.org/10.1007/s10586-024-04651-9
7. Kamble, Torana, Sanjivani Deokar, Vinod S. Wadne, Devendra P. Gadekar, Hrishikesh Bhanudas Vanjari, and Purva Mange. "Predictive Resource Allocation Strategies for Cloud Computing Environments Using Machine Learning." Journal of Electrical Systems 19, no. 2 (2023).
8. Khan, Ahmad Raza. 2024. "Dynamic Load Balancing in Cloud Computing: Optimized RL-Based Clustering with Multi-Objective Optimized Task Scheduling" *Processes* 12, no. 3: 519. https://doi.org/10.3390/pr12030519
9. Wang, Z., Chen, S., Bai, L. et al. Reinforcement learning based task scheduling for environmentally sustainable federated cloud computing. J Cloud Comp 12, 174 (2023). https://doi.org/10.1186/s13677-023-00553-0
10. Zhang, Y., Wang, J. Enhanced Whale Optimization Algorithm for task scheduling in cloud computing environments. J. Eng. Appl. Sci. 71, 121 (2024). https://doi.org/10.1186/s44147-024-00445-3
11. Sanjay, N., & Sreedharan, S. (2024). Hybrid Bio-Inspired Optimization-based Cloud Resource Demand Prediction using Improved Support Vector Machine. International Journal of Advanced Computer Science & Applications, 15(1). 10.14569/ijacsa.2024.0150177